Carrier Recovery

Joe Leacy

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1 Introduction

The performance and energy consumption of several combinations of frequency and phase recovery algorithms are compared. CORDIC will be used to find angles and approximated as 3n additions. Direct complex multiplication used to apply phase rotations. Bit shifts will be assumed to have negligible energy cost.

2 Frequency Recovery

2.1 Isolating CFO

All frequency recovery algorithms first require that the phase of the data be removed before estimation. For QPSK symbols $c[i] = e^{j\frac{\pi}{4}+m[i]\frac{\pi}{2}}$ this can be done by raising each received symbol $x[i] = c[i]e^{j(2\pi f_d i+\theta[i])} + n[i]$ to the fourth power to give

$$z[i] = x^4[i] = e^{j(8\pi f_d i+4\theta[i])} + n'[i] \tag{1}$$

where θ is phase noise, f_d is the CFO, and n is additive noise. This allows all data symbols to be used in CFO estimation, but ampifies phase and additive noise which degrades the estimation quality. The fourth power operation also requires three complex multiplications. An alternative is to use pilot symbols where the data phase is known beforehand, giving

$$z[i] = x[i]c^*[i] = e^{j(2\pi f_d i+\theta[i])} + n[i], \tag{2}$$

which is more robust to noise and less computationally expensive. Some algorithms only require $\arg(z[i])$ which is efficiently found using CORDIC as

$$\arg(z[i]) = \{4\arg(x[i])\}|_{-\pi}^{\pi} \tag{3}$$

for blind operation or

$$\arg(z[i]) = \{\arg(x[i]) - \arg(c[i])\}|_{-\pi}^{\pi} \tag{4}$$

for data-aided operation where the wrap operation can be done with no additional cost by aligning the wrap range to the overflow range of the fixed point value used to store the phase.

2.2 Rife and Boorstyn (FFT Search)

The value of the CFO will appear as the strongest peak in the frequency spectrum of $z[i], z[i+1], \ldots, z[i+L]$ where L is the observation length. The Rife and Boorstyn algorithm first performs an FFT on the observed sequence and finds the index k corresponding to the strongest peak. This coarse estimation is refined with a parabolic interpolation on the complex FFT values X_{k-1}, X_k, X_{k+1} to give

$$\delta = -\Re\left[\frac{X_{k+1} - X_{k-1}}{2x_k - X_{k-1} - X_{k+1}}\right]. \tag{5}$$

The updated index $k' = k + \delta$, which is then scaled by the sampling rate to find the true CFO. The CPON specification allows for 11 training symbols at the start of each subframe. The resolution in k is determined by the FFT size N_{FFT} as $\epsilon(k) = 1/2N_{FFT}$. For data-aided operation, only the 11 training symbols are available. This would give a resolution of $\epsilon(k) \approx 0.05$. In terms of the estimated frequency $\hat{f}_d$ and symbol rate R_s, $\epsilon(\hat{f}_d) \approx 0.05R_s$. This would likely be too large for the downstream phase recovery to correct, so a larger FFT can be used with zeros padded at the end of the observed sequence. The normalised mean squared-error (MSE) in the estimated CFO is shown in fig. 1 for data-aided operation with 11 training symbols. A CFO estimate is computed for both polarizations and averaged to reduce the MSE. The modified Cramer-Rao bound (MCRB) gives a lower bound on the achievable MSE and is given by

$$MCRB = \frac{3}{2\pi^2 L^3 \times SNR}. \tag{6}$$

The Rife and Boorstyn achieves this bound for SNRs over ≈ 4 dB (the threshold SNR). At lower SNRs, the noise has a significant contribution to the FFT values and can lead to an incorrect index being chosen.

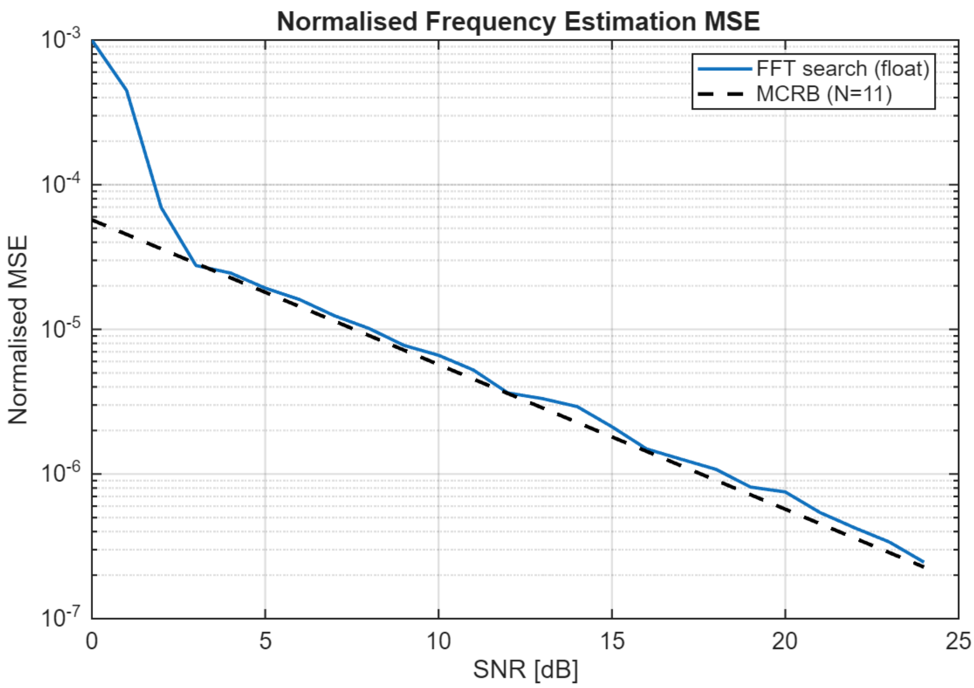

Figure 1: Normalised MSE of the Rife and Boorstyn algorithm with FFT size $N_{FFT} = 512$ compared to the MCRB for $L = 11$ training symbols.

Data-aided operation is constrained by the length of the training sequence but the accuracy of the Rife and Boorstyn algorithm improves with longer observation lengths, as shown in fig. 2 for non data-aided operation. The SNR threshold at which this algorithm begins to deviate from the MCRB is initially higher with data aided operation due to the amplification of noise but decreases for longer observation lengths.

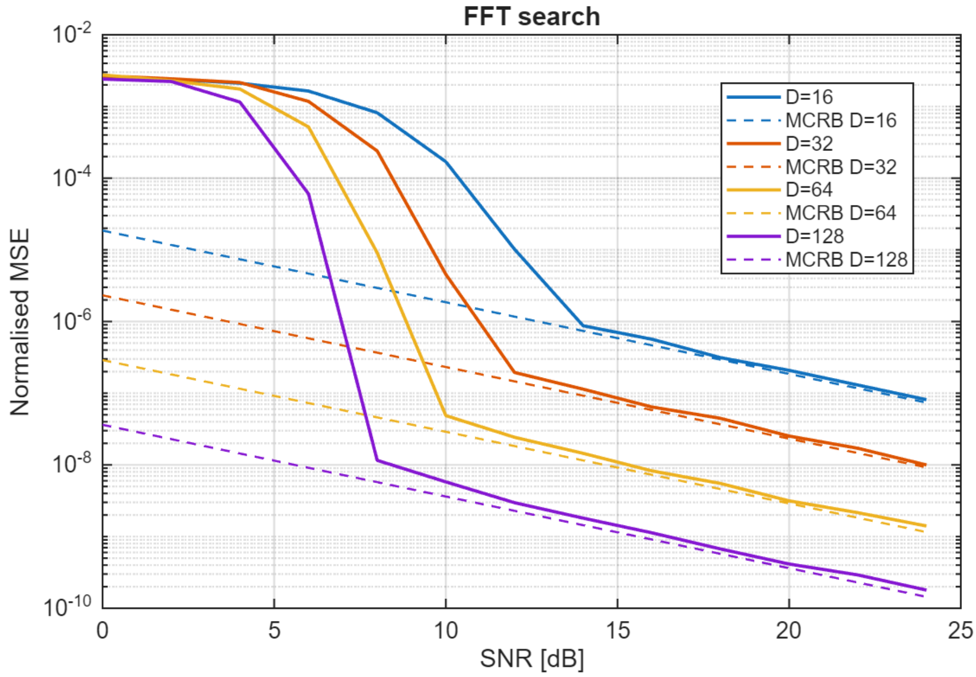

Figure 2: Normalised MSE of the Rife and Boorstyn algorithm for non data-aided operation and different observation lengths

2.3 Differential Phase + Kay Estimator

The Rife and Boorstyn algorithm exhibits good performance even at low SNRs but is computationally expensive. The FFT requires $\frac{N_{FFT}}{2}\log_2(N_{FFT})$ complex multiplications and $N_{FFT}\log_2(N_{FFT})$ complex additions. The fine search from eq. (5) also requires a complex division. A cheaper approach would involve computing the phase difference

$$\arg(z[i]) - \arg(z[i-1]) = 2\pi f_d T_s + \theta[i] - \theta[i-1] + \eta[i] \tag{7}$$

over several symbols and averaging to remove the effect of θ. Estimating f_d like this is vulnerable to the effect of additive noise on the phase $\eta[i]$ so exhibits poor accuracy as shown in fig. 3. This can be improve with a second pass using the Kay estimator. The Kay estimator uses a least-squares approach, computing the estimated CFO as

$$\hat{f}_d = \frac{R_s}{2\pi}\sum_{k=1}^{L}\frac{6k(L-k)}{L(L^2-1)}\arg(z[k]z^*[k-1]). \tag{8}$$

where $\arg(z[k]z^*[k-1])$ computes the phase difference from eq. (7) and $\frac{6k(L-k)}{L(L^2-1)}$ is a weighting term that prioritises the contribution of phase differences in the middle of the observed sequence. This estimator is able to achieve the MCRB but the least-squares approach considers phase differences of non-adjacent symbols, making it vulnerable to phase wrapping that distorts the estimate when the CFO is high enough. This is demonstrated by considering the phase of symbols at either end of the observation sequence. The phase drift between these symbols $\Delta\phi = 2\pi f_d L T_s$. If $|\Delta\phi| > 2\pi$ then a phase wrap is guaranteed to occur, giving an estimated phase drift $\Delta\hat{\phi} = \Delta\phi \pm 2\pi$ and an estimated CFO $\hat{f}_d = f_d \pm \frac{1}{LT_s}$. Thus the CFO correction from first pass is needed to bring the residual CFO within range for the Kay estimator.

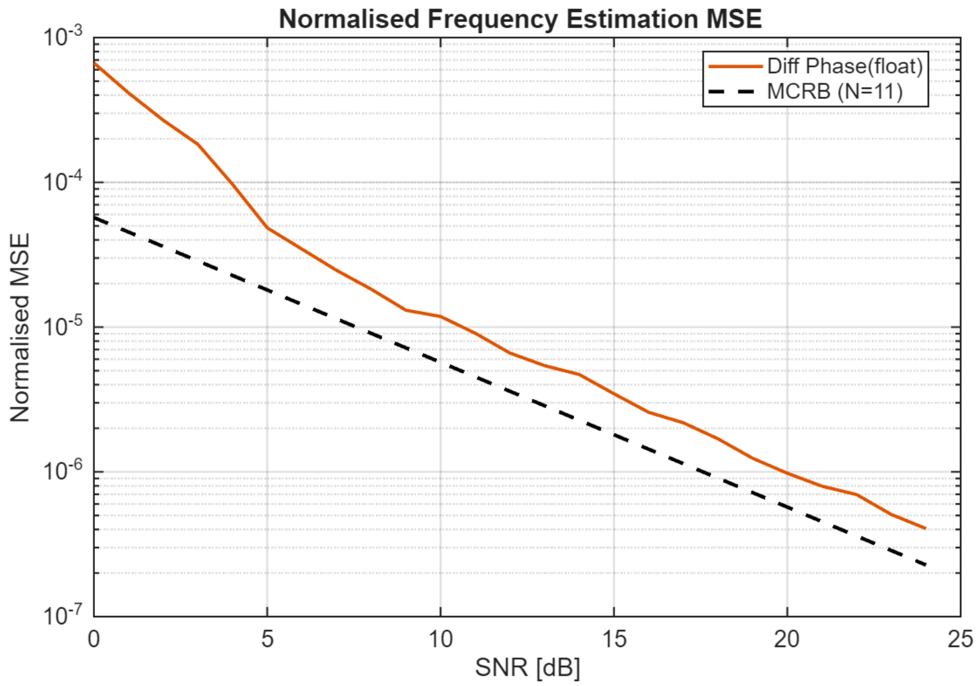

Figure 3: Normalised MSE of frequency estimation with data-aided differential phase for $L = 11$ training symbols.

The performance of the combined differential phase and Kay estimator for data-aided operation is shown in fig. 4. The MCRB is achieved above an SNR of ≈ 7 dB. This is higher than for Rife and Boorstyn but at a reduced computational cost. Unlike the Rife and Boorstyn algorithm, the threshold at which the MSE of this estimator deviates from the MCRB does not depend on obervation length, as shown in fig. 5.

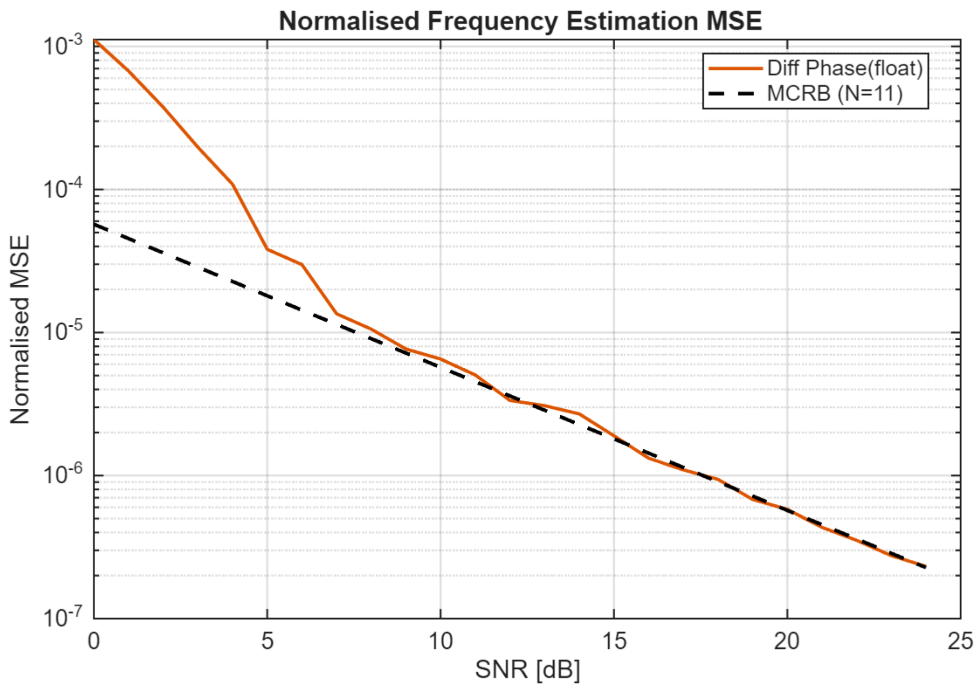

Figure 4: Normalised MSE of frequency estimation with data-aided differential phase and Kay estimator for $L = 11$ training symbols.

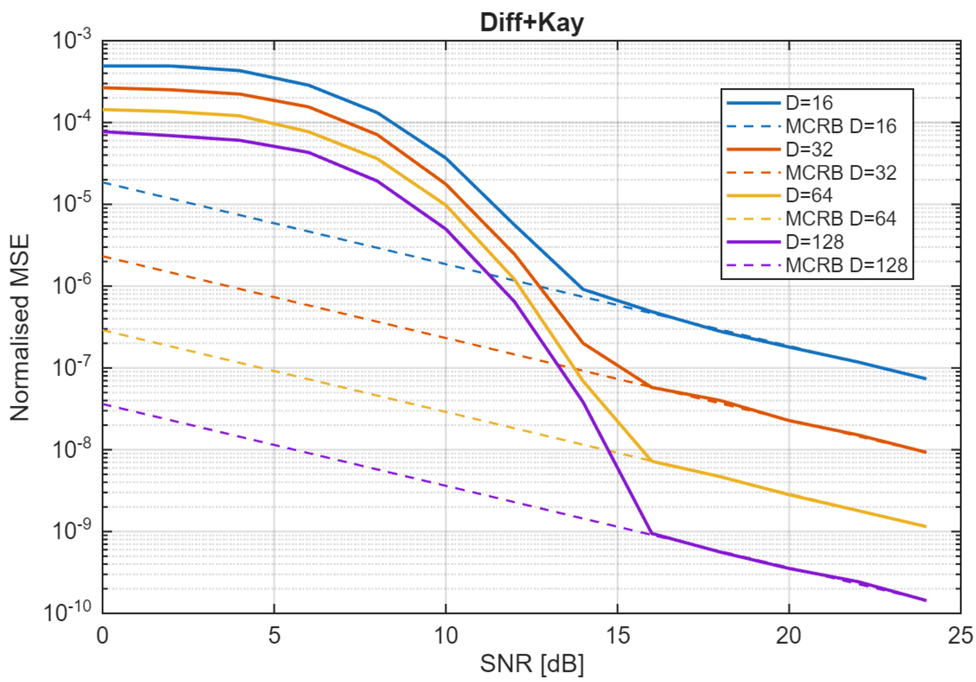

Figure 5: Normalised MSE of the differential phase and Kay estimator with blind operation for different observation lengths.

2.4 Computational Complexity

Here the computational complexity of both frequency recovery algorithms are compared in terms of real multiplications and real additions. CORDIC is used to find phases and is assumed to cost the same as a real multiplication. Real division will also be assumed to cost the same as multiplication as it is only performed once (this would be a suitable approximation if look-up tables are used to find reciprocals at the cost of accuracy). All analysis is for data-aided operation, avoiding the fourth power operation in eq. (1) and is only for a single polarization. If averaging is done over both polarizations then all operations are doubled.

2.4.1 Rife and Boorstyn

The Rife and Boorstyn algorithm requires an FFT followed by N_{FFT} comparisons of the magnitudes in the coarse search. Many of the FFT entries are 0 due to padding, which reduces the total number of operations needed. The interpolation requires the real part of a complex division

$$\Re\left[\frac{a+bj}{c+dj}\right]=\frac{ac+bd}{c^2+d^2}. \tag{9}$$

The total complexity is summarised in table 1 with the complexity for blind operation marked with (*).

Operation	RM	RA
Forming $z[i],\ldots,z[i+L]$	$4L$, $12L^*$	$3L$, $9L^*$
FFT	$2L\log_2(\frac{N_{FFT}}{L})+2N_{FFT}\log_2(L)$	$3L\log_2(\frac{N_{FFT}}{L})+3N_{FFT}\log_2(L)$
Search	$2N_{FFT}$	$2N_{FFT}$
Interpolation	5	4

Table 1: Computational complexity of estimating CFO with Rife and Boorstyn algorithm

2.4.2 Differential Phase + Kay

The proposed differential phase and Kay estimator is less complex, only requiring CORDIC, real additions, and real multiplications. The fixed multiplications by the weights of the Kay estimator can be optimised into a low-energy shift-add circuit but this is ignored for simplicity. The first pass CFO correction involves forming the phase corrections of each training symbol as $\phi[i]=\hat{f}_d i$, then applying them with CORDIC.

Operation	RM	RA
Forming $\arg(z[i]),\ldots,\arg(z[i+L])$	$2L$	L
Differential Phase	1	$L-1$
First Pass Correction	$2L$	0
Forming $\arg(z[i]),\ldots,\arg(z[i+L])$	$2L$	L
Kay	L	$L-1$

Table 2: Computational complexity of estimating CFO with the proposed differential phase and Kay estimator

2.5 Optimising Bit Width

These frequency recovery algorithms compute an estimate of the normalised CFO f_d/R_s. For n bits of precision, the smallest normalised CFO that can be estimated is 2^{-n} when

$0 \leq |f_d/R_s| \leq 1$ is the estimation range. In reality a much smaller range of CFOs needs to be estimated. If the maximum permitted value of $|f_d/R_s| = 2^{-d}$, d bits of precision that would otherwise be redundant can be utilised by scaling each estimate by 2^d during calculation then inserting d bits in the final answer to recover the true normalised CFO. This improves the resolution to $2^{-(n+d)}$ without any increase in energy consumption.

3 Phase Recovery

3.1 Feed-Forward vs PLL Algorithms

One of the first techniques investigated for phase recovery was a digital phase-locked loop (PLL) to incrementally correct the phase of each symbol with feedback. This dramatically simplifies the phase estimation and correction due to the use of small angle approximations. The CPON spec sets the symbol rate at 30.5 GBd. With an oversampling rate of 8/7 the total samples per second would be 35 GSa/s. This is beyond the achievable clock speed of CMOS, so symbols must be processed in blocks of e.g 32 symbols with a clock speed of 1.1 GHz. This approach hinders the effectiveness of a PLL, as updates can only be made every 32 symbols. For this reason, feed-forward algorithms that can operate on blocks of data were used instead.

3.2 Viterbi & Viterbi

The Viterbi & Viterbi algorithm is well suited to QPSK andremoves data phase with the fourth power operation of eq. (1). It then applies a maximum-likelihood filter $w_{ML}[i]$ to estimate the phase error due to Wiener phase noise from the combined laser linewidth and any residual CFO as

$$\hat{\theta}_{ML}[i] = \frac{1}{4} \arg\left(\sum_{i=1}^{N} w^*_{ML}[i] z[i]\right) - \frac{\pi}{4}. \tag{10}$$

A phase unwrapper is then applied before the phase error is corrected. This is readily adapted to block-based processing. If blocks of 32 symbols are used again, the ML filter can be applied on all symbols in the block and the estimated phase error applied to the entire block. This is only valid if the phase difference between the start and end of the block is small, meaning the laser linewidth cannot be too large and the preceding frequency recovery algorithm must correct most of the CFO. By increasing the time between phase estimates, block processing also increases the chance of cycle-slips in the phase uwnrapper. The CPON spec supplies 1 pilot symbol in every 32 symbol block that can be used to correct this [?]. The correction uses a reference phase error measured from the pilot, then adds or subtracts multiples of the period of the phase unwrapper if the difference between the estimated and reference phases exceeds a threshold. The thresholds that gave the lowest BER are shown in table 3 for low SNRs, where cycle slips are more likely to occur. A threshold in the range $90 - 112.5°$ seems optimal and doesn't change with SNR or linewidth.

SNR (dB)	100 kHz	1 MHz	10 MHz
0	90	90	90
2	90	90	90
4	90	90	90, 112.5
6	90	90	112.5
8	112.5	90	112.5
10	90	112.5	112.5

Table 3: Optimal threshold (°) for pilot-based cycle slip correction in the Viterbi & Viterbi algorithm for different combinations of SNR and laser linewidth, $R_s = 30.5$ GBd

3.3 Pilot-Only Correction

While Viterbi & Viterbi is commonly used for QPSK, a simpler approach would be to just compute the phase error from the pilot symbol using eq. (2) and apply this to the whole block. This also relies on a small phase change across the block and is more vulnerable to additive noise. However, QPSK allows for ±45° error in the phase estimate before an incorrect symbol decision is made which makes it robust to the drop in accuracy associated with this approach.

3.4 Computational Complexity

3.4.1 Viterbi & Viterbi

For a block-based approach, the length of the filter will be the block length. Phases are again found with CORDIC. The cost of the phase estimate for each block is shown in table 4.

Operation	RM	RA
Forming $z[i], \ldots, z[i+N]$	$12N$	$9N$
Computing $\hat{\theta}_{ML}$	$4N+1$	$3N+2(N-1)+1$

Table 4: Computational complexity of block-based Viterbi & Viterbi

3.4.2 Pilot-Only

Only phases are needed here so the entire estimation can be performed with CORDIC. The pilot phase is also known beforehand so doesn't contribute to the computational cost. The total cost of the phase estimate is shown in table 5

Operation	RM	RA
Forming $\arg(z[i]) = \hat{\theta}$	1	1

Table 5: Computational complexity of proposed pilot-only correction.

4 Combined Performance

The CPON specification requires the bit-error rate (BER) to be below 2×10^{-2}. The required SNR tp achieve this for each frequency-recovery and phase-recovery combination are shown in table 6. This was computed by averaging the BER over 100 trials for a range of SNRs, then finding the crossing point of the FEC limit by interpolation. All calculations were done with 16 bits of precision.

Scenario	R&B—VV	R&B—PO	DK—VV	DK—PO
DA	7.848	8.245	7.827	8.107
BL - 64	8.229	8.282	15.741	15.559
BL - 256	7.371	7.468	15.341	15.199
BL - 512	7.381	7.472	14.995	14.966

Table 6: SNR (dB) at FEC limit for each frequency-recovery and phase-recovery combination for data-aided (DA) operation and a range of blind (BL) observation lengths. $N_{FFT} = 512$.

4.1 Blind vs Data-Aided Operation

The lower threshold SNR of data-aided operation for both algorithms results in better performance than blind operation with 64 observations. However, the reduction in threshold SNR for the Rife and Boorstyn algorithm means that longer observation lengths (256 and 512) outperform the data-aided case. For the differential phase and Kay estimator, the threshold SNR does not decrease and the required SNR to reach the FEC limit remains high. *plot required SNR vs blind observation length?*

4.2 Algorithm Choice

For data-aided operation, the choice of phase recovery algorithm affects performance more significantly than the frequency recovery algorithm, with the DK estimator performing marginally better than R&B. The proposed pilot-only correction required on average 0.34 dB more of SNR than VV to achieve the FEC limit.

4.3 Bit Width

Scenario	R&B—VV	R&B—PO	DK—VV	DK—PO
DA	7.451	8.618	7.890	8.223
BL - 64	7.934	7.948	15.823	15.688
BL - 256	7.382	7.468	15.264	15.092
BL - 512	7.327	7.399	15.265	15.160

(a) n = 16, m = 8.

Scenario	R&B—VV	R&B—PO	DK—VV	DK—PO
DA	17.330	8.371	17.312	8.265
BL - 64	17.281	8.341	17.317	15.607
BL - 256	17.268	7.606	17.296	15.245
BL - 512	17.280	7.618	17.307	14.984

(b) n = 16, m = 4.

Table 7: Effect on performance from frequency recovery bit width (n) and phase recovery bit width (m).